

Early warning from near-misses

Surfacing emerging aviation-safety risk early — from public data

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Executive summary

Accidents are rare and *lagging*; the near-misses that precede them are common and arrive *first*. This study asks whether publicly available data can surface those precursors **before** a risk is widely recognised — and demonstrates that it can, on a worked example.

Using 15 years of ASRS incident reports, two independent detectors were built on a single principle — track each risk category's **share** of monthly reports (not raw counts) and flag a sustained, statistically anomalous rise. One detector reads the **structured taxonomy codes**; the other is a **zero-shot language model** reading the free-text narratives. Both moved early — one as a rising trend, one as a statistical fire:

Surface / runway “ground-conflict” near-misses began rising in 2021–2022 — roughly 12–18 months before the public inflection (the FAA's March 2023 emergency Safety Summit and the August 2023 *New York Times* close-calls exposé). The structured ground-conflict tag roughly doubled ($\approx 5\% \rightarrow 11\%$ of reports, raw count tripled); the narrative model independently fired on runway incursions in **June 2021**.

The rise survives the obvious challenges: it is in the numerator not the denominator, it persists after normalising for the post-COVID traffic rebound (the incursion *rate* ran $\sim 15\%$ above pre-pandemic and *severity* spiked), and it pre-dates the summit (ruling out awareness-driven over-reporting). External FAA/DOT-OIG figures confirm both the rise and its 2024 easing.

The deliverable is not an accident predictor. It is a **leading indicator** and an **operating model**: the same pipeline produces a ranked monthly watchlist of *what is emerging and where to look first* — a triage queue for human analysts, months earlier than the lagging record would allow.

1. Problem framing

How should an “early-warning system” for aviation safety be interpreted? Not as accident prediction. Accidents are (thankfully) rare, and a model trained on them learns from the lagging record. The tractable, useful problem is **detecting precursors trending upward before the adverse outcome is widely recognised — from data already held**.

This reframing drives every subsequent choice:

- **What to monitor:** *precursors* — voluntary near-miss reports — rather than accidents. ASRS is a national, public repository of exactly these, filed by pilots, controllers, and maintenance personnel, and rich in free-text narrative.
- **What counts as a signal:** a **sustained rise in the share** of reports about a given risk — a change in the *mix* of what is being reported, not merely in how much is reported.
- **What the output is:** a **ranked shortlist for human review**, explicitly framed as *leading indicators*, *not alarms* — a reason for an analyst to look, never an automated verdict.

This matches how the role’s domain actually operates: the value is in *framing, reasoning, and communication* of a credible early signal, not in a black-box predictor.

2. Data

Source. NASA ASRS, 2011–2025: **80,047** reports after de-duplication by accession number (`ACN`), **127** columns, of which **79,572** carry a usable free-text narrative. Combined from 15 annual public extracts.

Grain and limitations (stated up front, because they bound every claim):

Property	Reality	Consequence
Time	<code>Date</code> is YYYYMM only (month grain, 100% filled)	Monthly series are the finest temporal resolution; no exact-date or intraday analysis
Location	No airport / origin–destination (de-identified by design); only facility <i>type</i> and US state	No airport-level hotspotting; geography limited to coarse slices
Reporting	Voluntary ; total monthly volume swings (2019 peak, 2020 COVID trough)	Raw counts conflate “how much is reported” with “what happened” → drives the share-based metric (§3)
Latency	~60-day processing lag	The most recent 2–3 months are under-reported and excluded from “fired” claims (shown but greyed)

Well-populated signal carriers include `Anomaly` (99.9%, multi-label taxonomy of what went wrong), `Contributing Factors / Situations`, `Person 1 | Human Factors`, `Aircraft 1 | Flight Phase` (96.7%), and the `Report 1 | Narrative` free text (100%). Multi-label fields are `" ; "`-joined and exploded so a report contributes to every category it lists.

A QA-first exploratory pass (`production/build/01-eda.ipynb`) established schema integrity, temporal coverage, missingness, and reporting-bias diagnostics before any modelling.

3. Method — the operating model

The architecture is deliberately simple and transparent: **one detection engine, fed by two independent labellers.**

3.1 One detection engine (shared)

Every label — whether a structured tag or a narrative-derived label — is turned into a monthly series and run through identical logic:

1. **Metric — share, not counts.** $s_c(t) = \text{count}_c(t) / \text{total_reports}(t)$. A signal is a shift in *share*, which controls for overall reporting propensity (the central ASRS caveat).
2. **Robust anomaly score.** Against a trailing **24-month** baseline, $z = (s_c(t) - \text{median}_{24}) / (1.4826 \cdot \text{MAD}_{24})$. Median/MAD (not mean/SD) so the baseline is not inflated by the very spikes being hunted.
3. **Thresholds + persistence.** $|z| \geq 2 = \text{watch}$, $\geq 3 = \text{strong}$; a fire requires the threshold to hold for **≥ 3 consecutive months** (kills one-off blips). Early-warning focuses on rising shares.
4. **Bias guards (mandatory before a fire counts).**
 - **Taxonomy-change guard** — categories whose first non-zero month is after 2011 may reflect a code being introduced/renamed; they are gated from signal claims.
 - **Seasonality guard** — a fire requires a positive year-over-year share change, so a recurring seasonal bump alone will not trigger.
 - **Multiple-testing stance** — with ~ 40 categories $\times \sim 180$ months, outputs are framed as *ranked candidates* ($\text{rank} = \text{peak}_z \times \text{months_sustained}$), not alarms.

3.2 Two labellers (differ only in how a report is labelled)

- **Structured view** — labels come from the ASRS taxonomy codes (*Anomaly* , *Flight Phase* , ...). Precise, but only sees risks that already carry a code.
- **Narrative view** — a **zero-shot LLM** reads the free text and assigns ~ 10 plain-English risk labels (e.g. *runway incursion / ground conflict*, *GPS interference / jamming*, *unstabilized approach*, *fatigue*). It catches the **emerging and unlabelled** — risks with no taxonomy code at all. A second, **unsupervised** topic-discovery pass (embeddings $\rightarrow$ clustering) guards against the “only finds what you name” confirmation-bias risk.

The strongest signal is one where **both views agree** — a narrative theme rising in step with a structured category — because they are entirely separate pipelines with different blind spots.

Implementation note (honest record). The narrative labeller was *prototyped* on a local GLiNER2 model (free, CPU). At full coverage that path measured ~ 66 h on CPU for 80k narratives, so the production run used a hosted zero-shot model — **OpenRouter deepseek/deepseek-v4-flash**, concurrent batched calls, across all **79,572** narratives. Only the labelling engine changed; the §3.1 detector logic is identical. (Output files retain a

4. Finding — surface & runway near-misses rose early

The structured detector surfaced **42 candidate series (26 with a persisted fire, 0 gated as taxonomy artifacts)**. The locked headline — chosen because it is falsifiable and externally checkable — is **surface / ground-conflict near-misses**, where a rising *structured trend* and a *persisted narrative fire* coincide.

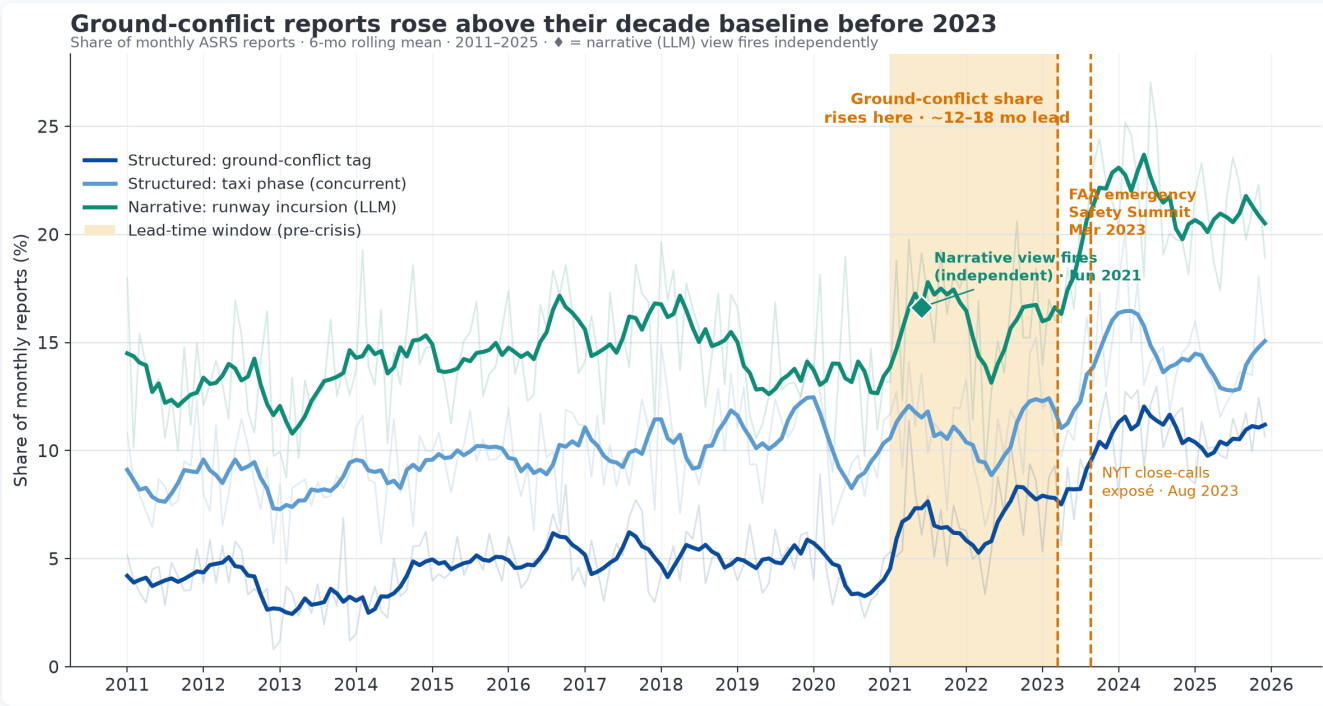


Figure 1 — Structured ground-conflict and taxi-phase shares with the independent narrative runway-incursion line (♦ marks the narrative fire, June 2021); the amber band is the pre-crisis lead-time window.

The structured **Conflict Ground Conflict** share roughly doubled:

Year	2011	2013	2021	2023	2024	2025
Ground-conflict share	4.1%	3.0%	6.7%	9.0%	11.2%	10.5%

corroborated by the concurrent **Taxi**-phase rise (~8–11% → 15.2% in 2024). **Independently, the narrative model fired on the runway incursion / ground conflict / surface movement label in June 2021** (persisted watch fire). Two methods, different pipelines, same trend and timing.

The most instructive detail: the structured share **spiked above the alarm threshold repeatedly** — its robust z reached ≈6, including in April 2021 — but **never for three consecutive months** (its longest watch run was 2), so the persistence rule never let it register a *fire*. The noisier structured series kept tripping and resetting; the **narrative** label rose smoothly enough to hold the 3-month rule

and produced the **only sustained fire — June 2021**. Two detectors, two different failure modes — the argument for running both, not a weakness to hide.

4.1 It led the publicly-recognised crisis by ~12–18 months

When	What	Source
Jun 2021	Narrative detector fires on runway incursions	this analysis
2021–2022	Structured ground-conflict share inflects upward	this analysis
Jan / Feb 2023	JFK (AA777 runway crossing) and Austin (SWA/FedEx ~150 ft) near-collisions	NTSB / press
Mar 2023	FAA convenes emergency Aviation Safety Summit	FAA newsroom
Aug 2023	NYT: “Airline Close Calls Happen Far More Often Than Previously Known”	NYT

The ASRS-derived signal **led the widely-recognised crisis by roughly 12–18 months**. Notably, the FAA’s own response was to *mine its ASIAS safety database for emerging trends* — **reactively, after** the events. The same idea, applied **proactively**, is exactly what this method demonstrates.

5. Robustness — ruling out the artifacts a reviewer will probe

5.1 Is the rising *share* just a denominator / compositional artifact?

No. Checked directly against the count series:

Year	Ground-conflict count (numerator)	Total reports (denominator)	Share
2011	230	5,646	4.1%
2021	305	4,571	6.7%
2023	411	4,589	9.0%
2024	606	5,427	11.2%
2025	642	6,109	10.5%

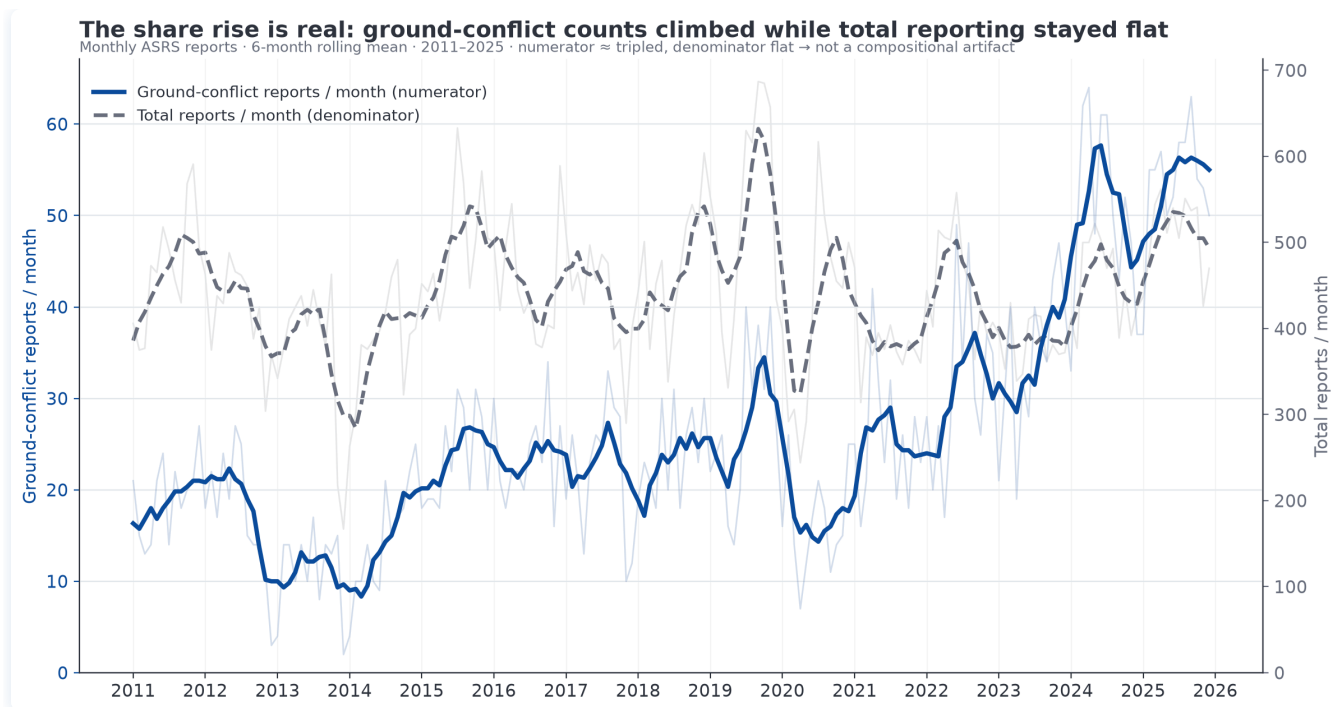


Figure 2 — Ground-conflict raw count (numerator) vs total monthly reports (denominator): the numerator roughly tripled while the base stayed flat.

The **numerator roughly tripled** while the **denominator stayed in a flat ~4,500–6,100/yr band**.

Because **Anomaly** is **multi-label**, category shares do not sum to 100% — one category rising does not mechanically force another down, so the naive pie-slice artifact does not apply. External absolute counts rose independently.

5.2 Isn't it just the post-COVID traffic rebound?

The single biggest challenge — *more flying* ⇒ *more conflicts, mechanically* — is ruled out on two counts:

- **The rate rose.** FY2023 had **fewer operations than FY2019 (37.3M vs ~43M)** yet **more incursions** (~1,777, ~5/day) → incursions-per-operation ran **~15% above** the pre-pandemic baseline.
- **Severity, not just frequency, spiked.** Category A&B (“serious”, collision narrowly avoided) incursions hit **22 in FY2023 — a six-year high — then fell to 9 in FY2024 (–59%)**. More traffic raises frequency, not severity; this mirrors the share peaking ~2024 and easing in 2025.

5.3 Other guards

- **Reverse causality** (post-summit awareness → more reporting): the signal **pre-dates** the March 2023 summit, so awareness-driven over-reporting cannot explain the leading edge — it strengthens the claim.
- **Taxonomy artifact:** `Conflict Ground Conflict` exists from 2011 (`taxonomy_suspect = False`).
- **Seasonality:** fires require positive YoY change.

- **Units caveat:** external counts are fiscal-year, raw; the figures here are calendar-month share — directionally consistent, *not* like-for-like (stated explicitly).

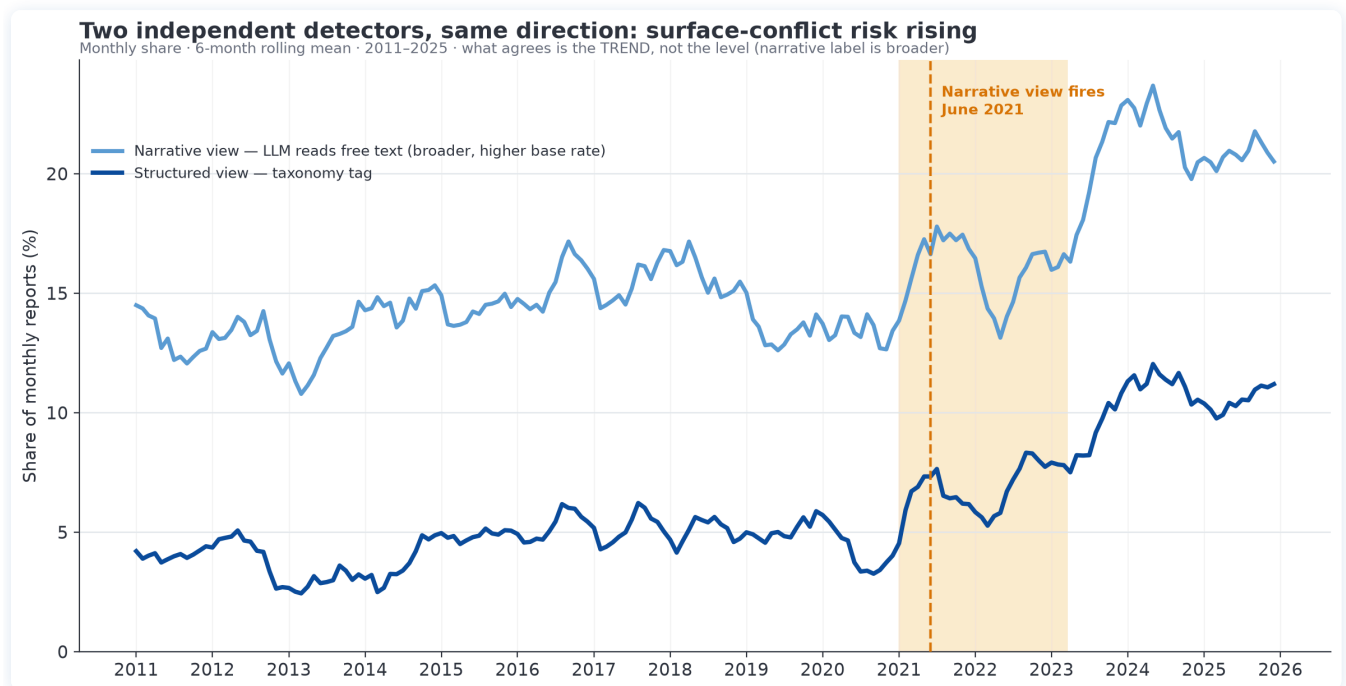


Figure 3 — Two independent detectors on one axis: the structured taxonomy tag and the narrative LLM both rise into 2021–24 (agreement in direction and timing, not level).

5.4 Does the z-threshold assume normality? (distributional check)

The robust z carries a normal-theory reading ($z \geq 2 \approx$ a 1-in-20 deviation), so the share distribution was checked directly (`make_normality_diagnostic.py`). It is **not** normal: the ground-conflict share is right-skewed (Shapiro–Wilk $p \approx 3 \times 10^{-8}$), and the pooled robust- z across all 42 candidate categories is fat-tailed (excess kurtosis +5.1) — empirically $|z| \geq 3$ occurs in **4.2%** of category-months versus the **0.27%** normality would imply ($\sim 15\times$). This does **not** undermine the method: median and MAD assume no distribution, so the detector stays valid. The consequence is interpretive — the 2/3 cut-offs are used as an **empirical screening rank, not calibrated p-values** (no significance level is ever claimed) — and the fat tails are precisely *why* the **persistence rule**, not a p-value, is the real false-positive control: a single extreme month is common, so a signal must hold for three.

Distributional diagnostic — the share series is non-normal; robust z is a screening rank, not a calibrated p-value

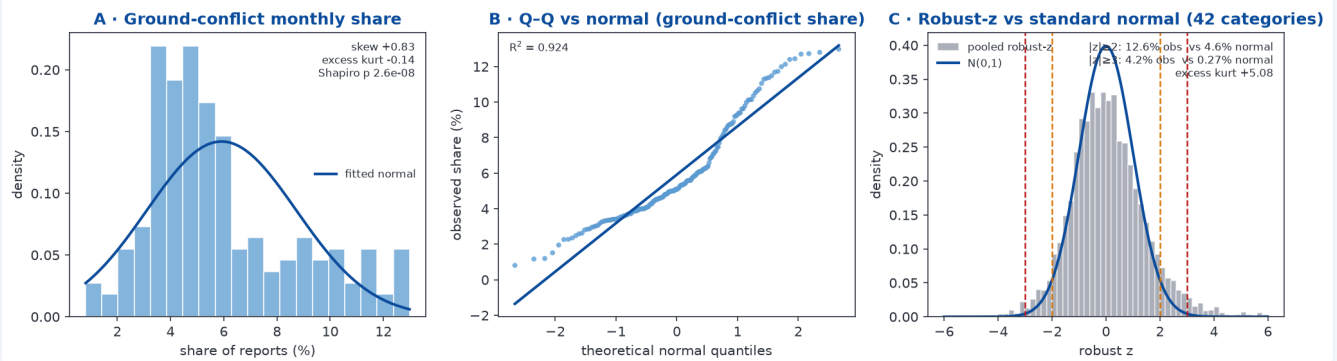


Figure 4 — Distributional diagnostic. The ground-conflict share is non-normal (A: right-skewed histogram; B: Q-Q tails bend off the line), and the pooled robust-z is markedly fat-tailed versus $N(0,1)$ (C) — so the threshold is a robust screening rank, not a significance test.

6. Validation — was the language model checked?

The narrative LLM is the novel half of the method and carries the formal June-2021 fire, so its output was spot-checked. **30 narratives the model flagged** with the runway label were independently adjudicated (positives only → precision, not recall; confidence-stratified, seeded for reproducibility — `production/build/validate_narrative_labels.py`):

Metric	Value
Precision — broad (genuinely surface/ground-related)	70% (21/30)
Precision — strict (true incursion/conflict near-miss)	37% (11/30)
Precision by confidence ≥ 0.95 / 0.90 / ≤ 0.85	88% (7/8) / 70% (7/10) / 58% (7/12)

The confidence bands are small ($n = 8 / 10 / 12$), so read the gradient as directional rather than precise — the takeaway is simply that model confidence is informative.

Dominant failure mode: 8 of 9 false positives are *airborne* near-mid-air collisions the model tags as “conflict” — the word “conflict” in the label bleeds airborne events into a surface label. A confidence threshold (~ 0.95) or an airborne-vs-surface guard would sharpen it materially.

Why the headline still holds: the label is broad (it flags $\sim 15.7\%$ of narratives; base rate $\sim 13\%$ even in 2011), so its *absolute level* is overstated. The early-warning claim therefore rests on the **trend** (the rise and the June-2021 fire) and on **agreement with the cleaner structured tag** — which does not share this failure mode — *not* on the narrative level. A 70%-broad / 37%-strict precision is acceptable for a triage shortlist; it would not be acceptable for an automated count, which is precisely why the output is framed as a candidate for review.

7. From one signal to a monthly watchlist (decision-support)

This worked example is a **proof of concept**. The same pipeline already emits a **ranked monthly watchlist** from both detectors — a short, explainable list of what is emerging and where to look first.

Monthly watchlist — ranked, explainable shortlist from both detectors

Candidates for human review, not alarms · 'peak z' = strength of the share anomaly · status from the persistence rule

Risk signal	View	First fire	Peak z	Status / note
GPS interference / jamming	Narrative	2019-05	27.0	watch
Runway / ground conflict	Both views	2021-06	4.2	watch · headline · structured trend agrees
Smoke / fire / fumes / odor	Both views	2019-05	5.3	watch
UAS / drone encounter	Narrative	2014-07	6.2	watch
Unstabilized approach	Narrative	2014-02	4.8	strong
Crew / ATC workload strain	Structured	2022-04	5.2	strong · post-COVID; reverting 2025

Figure 5 — Example ranked monthly watchlist produced from both detectors: a short, explainable, prioritised queue for human triage.

Value to a safety team: instead of waiting for the lagging accident/incident record, an analyst gets a prioritised, *explainable* queue of rising precursors — each traceable to the reports behind it — months earlier. It supports decisions about **where to focus limited inspection, audit, and outreach attention**.

How it would slot into practice: - **Monthly run** on the latest ASRS extract; the freshest 2 months flagged soft (reporting lag). - **Human-in-the-loop triage** — candidates, not alarms; an analyst confirms or dismisses, building a feedback record. - **Cross-source corroboration** for anything that fires, exactly as done here.

8. Limitations — what this is, and what it isn't

What it is: a leading indicator that risk is *rising*; low-cost, public, explainable, reproducible; corroborated three ways (narrative fire + structured trend + external sources).

What it isn't: - **Not accident prediction** — these are precursors, not accidents. - **Not causal**. The leading hypothesis is post-COVID understaffing plus the traffic rebound — but only the *symptom* is observed, not the cause. - **Not bias-free**. Voluntary-reporting bias is *mitigated* (share, not counts) — not eliminated. - **Coarse**. Monthly, US-only, de-identified (no airport). These figures and the FAA's agree in *direction*, *not magnitude*.

9. Reproducibility

Component	Location
EDA / QA	production/build/01-eda.ipynb
Signal A (structured)	production/build/02-structured-signals.ipynb
Signal B (narrative LLM + discovery)	production/build/03-narrative-signals.ipynb
Method spec (single source of truth)	production/spec/methodology.md
External corroboration & bias rule-outs	research/notes/phase5-corroboration.md
Narrative validation	research/notes/narrative-validation.md
Distributional / normality diagnostic	production/build/make_normality_diagnostic.py → output/eda/normality_diagnostic.png
Figures & chart builders	production/output/ , production/build/make_*.py
Presentation	writing/deck.pdf

- **Data** is not committed (NASA ASRS public extracts); see the README for the documented download steps.
- **Environment:** Python 3.12, requirements.txt (Polars, pandas, scikit-learn, matplotlib, httpx).
- **Narrative model:** OpenRouter deepseek/deepseek-v4-flash (needs OPENROUTER_API_KEY); the derived label parquets are committed so Signal B is inspectable without re-running the paid step.

Sources

- NASA ASRS Database Online — asrs.arc.nasa.gov
- DOT OIG, “FAA Has Taken Steps To Prevent and Mitigate Runway Incursions” (final report, Mar 2025) — oig.dot.gov
- FAA, “Readout from the FAA Aviation Safety Summit” (Mar 2023); FAA Air Traffic by the Numbers (2024); FAA Runway Safety Statistics
- NYT, “Airline Close Calls Happen Far More Often Than Previously Known” (Aug 21, 2023)
- Honeywell Aerospace (serious incursions 16 → 23, 2022 → 2023)

Authorship & provenance: this analysis was produced with AI-agent assistance under human direction; every material contribution is logged in provenance-ledger.md with author, timestamp, and change. The transparency is deliberate — the method, the self-corrections (e.g. the structured z-detector staying silent), and the model pivot are all part of the record.